

Ashwin V George

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Education

Master of Science in Astrophysics

UNIVERSITY OF POTSDAM

Potsdam, Germany

06/2023 - 12/2026 (Expected)

- Master's thesis: Carving Cavities in the Cosmos: Simulations of AGN jets in Galaxy Groups and Clusters
- Current average: **1.5**

Bachelor of Science in Physics

MADRAS CHRISTIAN COLLEGE

Tamil Nadu, India

06/2019 - 06/2022

- Graduated with First Class honors

AISSCE All India Senior School Certificate Examination

MAR DIONYSIUS SCHOOL

Kerala, India

03/2018

- Average: **88.6%** (Physics: **95%**, Maths: **95%**)

Research Experience

Master Student

LEIBNIZ INSTITUTE FOR ASTROPHYSICS

Potsdam, Germany

05/2025 - Present

- Conducting master's thesis research on AGN jet feedback in galaxy groups and clusters, supervised by Dr. Rainer Weinberger.
- Running hydrodynamic simulations of resolved AGN jets using **Arepo**.
- Analysed halos from IllustrisTNG to validate initial conditions of the idealised setup.

Intern

MAX PLANCK INSTITUTE FOR ASTROPHYSICS

Garching, Germany

01/2025 - 05/2025

- Developed and executed **Athena++** based hydrodynamical simulations of cold gas clouds in the intracluster medium.
- Investigated survival criteria of cold gas clouds in a turbulent, multiphase medium, under the supervision of Dr. Max Gronke and Hitesh Das; results published in Research Notes of the AAS ([arXiv:2507.20962v2](#)).

Publications

Survival of higher overdensity cold gas in a turbulent, multiphase medium

Research Notes of the AAS

ASHWIN VERGIS GEORGE, HITESH KISHORE DAS, MAX GRONKE

2025

DOI: [10.3847/2515-5172/ae1e79](https://doi.org/10.3847/2515-5172/ae1e79), [arXiv:2507.20962v2](#)

Scholarships

DAAD Degree Completion Scholarship

GERMAN ACADEMIC EXCHANGE SERVICE (DAAD) – UNIVERSITY OF POTSDAM

Potsdam, Germany

10/2026 - 02/2027

- Selected by a university committee based on academic performance, extracurricular engagement, and skills acquired beyond the field of study.
- Funded by the German Federal Foreign Office.

Additional Research Experience

Student Research Assistant

DEUTSCHES GEOFORSCHUNGSZENTRUM

Potsdam, Germany

12/2024 - Present

- Developing and maintaining Python pipelines to process auroral imagery and magnetometer data, working with Dr. Jürgen Matzka.
- Providing remote monitoring and technical support for various instruments at Neumayer Station III, Antarctica.

Student Research Assistant

HELMHOLTZ-ZENTRUM BERLIN FÜR MATERIALIEN UND ENERGIE

Berlin, Germany

09/2024 - 12/2025

- Developed micromagnetic simulations to model spin-wave propagation and skyrmion dynamics in magnetic materials with Dr. Steffen Wittrock.
- Ran GPU-accelerated simulations using BORIS Computational Spintronics.

Beamtime User at FERMI Free Electron Laser

Trieste, Italy

ELETTRA-SINCROTRONE TRIESTE

10/2024

- Part of an experiment to study magnons and their propagation on different materials. Contributed to data collection, analysis and experiment control of the FEL.

Student Research Assistant

Berlin, Germany

HELMHOLTZ-ZENTRUM BERLIN FÜR MATERIALIEN UND ENERGIE

08/2024 - 11/2024

- Designed a GUI for the real-time monitoring of low-frequency radio wave signals in superconducting cavities, visualizing forward and transmission signals.

Conferences

Thinkshop 2025

Potsdam, Germany

LEIBNIZ INSTITUTE FOR ASTROPHYSICS

07/2025

- Week-long international conference on feedback in galaxy formation.
- Served on the Local Organizing Committee (LOC).

Schools

WE-Heraeus and NARIT Cosmology School 2025 – Galaxies and Beyond

Chiang Mai, Thailand

NATIONAL ASTRONOMICAL RESEARCH INSTITUTE OF THAILAND

10/2025

- Selected participant for a week-long international training program covering cosmology, galaxy formation, cosmological simulations, and the Λ CDM model. Funding awarded by the Wilhelm and Else Heraeus Foundation and NARIT to support travel, accommodation, and participation.

Jürgen Ehlers Spring School on Gravitational Physics

Potsdam, Germany

MAX PLANCK INSTITUTE FOR GRAVITATIONAL PHYSICS

02/2025

- Selected participant for spring school for introduction to gravitational physics, general relativity, black holes and gravitational-wave astrophysics.

ESCAPE Summer School

Remote

LABORATOIRE D'ANNECY DE PHYSIQUE DES PARTICULES

06/2021

- An online summer school for training in data science and practices relevant for astrophysics and particle physics.

Code/Astro

Remote

HEISING-SIMONS FOUNDATION

06/2021

- An online workshop conducted by astronomers from Caltech to teach fundamental software skills and best practices for building open source packages for astronomy applications.

ISSAA 2021

Remote

INTER-UNIVERSITY CENTRE FOR ASTRONOMY AND ASTROPHYSICS

05/2021

- The Introductory Summer School in Astronomy and Astrophysics is a summer school intended as a month-long introduction to the various sub-fields of astronomy and astrophysics.

Skills

Programming

Python (NumPy, SciPy, Matplotlib), C/C++, MATLAB, Bash

Simulation codes

AREPO, ATHENA++, BORIS Computational Spintronics

Tools

SLURM job management on HPC clusters, Git, LaTeX, Linux

Online courses

Machine learning and data analysis (Coursera, Kaggle, freeCodeCamp)

References

Dr. Rainer Weinberger

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Leibniz Institute for Astrophysics
Potsdam, Germany
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Dr. Max Gronke

ERC Research Group Leader
Astronomisches Rechen-Institut
Heidelberg, Germany
max.gronke@uni-heidelberg.de